
Shared-Ride Platform

Scheduled shared transport for Alexandria, Egypt

Information document describing the product design, operating model, revenue mechanics, evidence base and known risks.

Document	Product and operating design
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Status	Pre-development. Specification complete.
Prepared from	Master Specification v3.0 (167 recorded decisions)

This document describes work completed to date and design intent. It contains no revenue history, no user numbers and no financial projections, because none exist. Figures drawn from third-party research are attributed at the point of use and listed in Appendix B.

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1 Summary

A scheduled shared-transport service in which an operator defines fixed routes and departure times, drivers claim the departures they wish to work, and passengers buy a seat at one flat price per route.

1.1 What the service is

A passenger opens the application, selects a route, selects a boarding point on that route, and selects a departure time. They pay a single fixed price regardless of how far along the route they travel, board by presenting a code, and leave the vehicle wherever they choose along the way.

The operator defines the route network: which stops exist, in what order, at what price, and during which hours. The operator also publishes a grid of departure times. Drivers do not design routes or set prices; they select the departures they want to run, in two steps.

In structure this is a microbus network operated through software. The organising idea is not to replace the informal shared transport that Egyptians already use, but to make it schedulable, priced in advance, and accountable.

1.2 The four design choices that shape everything else

ONE

Passengers buy a route, not a destination.

A ticket grants travel on a route. The passenger boards at a defined stop and alights wherever they wish. No destination is declared at purchase. This mirrors existing microbus behaviour and removes a large amount of complexity from pricing and matching.

TWO

One flat fare per route.

The same price applies whether the passenger travels two stops or twelve. The price is fixed when the ticket is bought and does not change afterwards. There is no demand-based pricing.

THREE

Passengers walk to a stop.

Boarding happens at defined meeting points rather than at the passenger's door. Published industry figures indicate that walk-to-a-point services sustain discounts of roughly 25–40 % against a private ride, while door-to-door shared services sustain roughly 15–22 %. The walk is what makes the price difference possible.

FOUR

The operator schedules; drivers choose when to work.

Routes, fares and timetables are centrally defined. Drivers claim published departure slots. They carry no planning burden and take no pricing decisions.

1.3 Current status

Design specification	Complete. 167 recorded decisions, 22 chapters, 75 defined screens, 65 configuration parameters, 34 system invariants.
Software	Not started. No code has been written.
Operations	Not started. No routes mapped, no drivers recruited.
Revenue	None. The service does not operate.
Regulatory position	Referred to legal counsel. Not assessed in this document.

This document therefore describes a design, not a business in operation. Where evidence exists it is external research, cited at the point of use. Where something is unknown, this document says so.

1.4 Why the design took this shape

The specification began as an on-demand pooled-ride concept and changed direction twice, in both cases because the available evidence pointed elsewhere.

- 1. Towards scheduled routes.** A 2026 evaluation framework covering fourteen studies published between 2007 and 2025 reports that fixed-route transit outperforms demand-responsive transit on cost per passenger, schedule adherence and ridership wherever demand is dense and regular, while demand-responsive services perform better in low-density areas. Published cost comparisons are wide: one North American agency recorded roughly \$72 per passenger on a flexible service against roughly \$25 on the fixed route it replaced. The intended launch context — university corridors at peak commuting hours — is the condition under which fixed routes perform best.
- 2. Towards recurring revenue.** Swvl, founded in Cairo in 2017 with a comparable consumer-facing model, moved away from consumer mass-market operation and reported reaching profitability in 2025 on business and government contracts with a high proportion of recurring revenue. Subscriptions are consequently central to this design rather than an accessory to it.

Source: full citations in Appendix B.

2 The problem and the market

2.1 How people currently travel

Alexandria’s shared transport is dominated by informal minibuses. They are cheap and frequent, and they already establish the behaviour this service depends on: people accept sharing a vehicle, accept walking to a boarding point, and accept a fixed fare for a known line.

What the informal system does not offer is a guaranteed seat, a known departure time, a recorded payment, an identified driver, or any means of complaint. Those five absences are the whole of the opportunity. This is a service-quality proposition, not a new mode of travel.

2.2 Why universities first

Dense demand	Shared transport requires demand density to function. Where demand is thin, vehicles run empty and the economics fail.
Predictable timing	Students travel at the same hours on the same days, which is the condition scheduled routes are designed for.
Price sensitivity	The population most likely to change behaviour for a lower fare.
Concentrated destinations	Campus gates create natural route endpoints.
Word of mouth	A closely connected population in which a service spreads or fails quickly and visibly.

2.3 Market sizing

NOT CLAIMED

No total addressable market figure is presented. A credible bottom-up estimate requires corridor demand data that has not been collected. Producing a top-down number from national population statistics would not inform any decision, and is therefore omitted. The measurements required to build a defensible estimate are listed in Chapter 7.

2.4 Comparable operators

Operator	Model	Relevance
Swvl	Scheduled buses and vans, seat booking, fixed routes	Cairo-founded, closest comparable. Consumer mass-market was loss-making; profitability reported in 2025 on recurring B2B and B2G contracts.
Uber, Careem	On-demand private and pooled rides	Different price tier. Operate under Egypt's ride-hailing regulation.
Informal minibuses	Fixed lines, fixed fares, no booking	The actual incumbent and the real competitor. Cheap and frequent; no guarantee, no record, no accountability.
Via, Lyft (international)	Walk-to-corner shared rides	Source of the pricing evidence used in this design. Confirms walk-to-a-point sustains materially deeper discounts.

3 How the service works

3.1 The passenger

1. **Choose a route.** Routes are named the way people already speak about them, for example Smouha to Alexandria University. The fare and frequency are shown alongside.
2. **Choose a boarding point.** Stops on the route are listed nearest first, with the walking time shown honestly. A more expensive option to be collected from the passenger's own street is offered alongside.
3. **Choose a departure.** At most three options are shown, each on a single line: departure time, boarding point, timetable arrival, price.
4. **Travel.** The passenger walks to the stop, boards by presenting a code, and leaves the vehicle at any point along the route.

The interface never asks which product the passenger wants. It asks where they are going and when. Internal vocabulary does not appear on screen.

3.2 The driver

A driver browses the published departure slots on approved routes and claims the ones they want. Two steps. They do not design routes, set prices or forecast demand.

This division was chosen deliberately. Requiring drivers to plan routes and predict demand imposes unpaid work before any earnings, which is a recognised cause of supply failure in peer-to-peer transport. A study of an in-house microtransit service found that the flexibility drivers valued was choosing *shifts*, not designing service.

3.3 The operator

Route network	Which stops exist, in what order, at what fare, during which hours.
Departure grid	Published slots, for example every fifteen minutes between 06:00 and 10:00.
Coverage	Visibility of unclaimed slots where passengers are waiting, with the ability to offer a bonus for claiming one.
Verification	Manual review of every driver: national identity document, driving licence, vehicle licence, photograph.
Support	Complaint handling, incident response and an on-call safety line.

3.4 Meeting points

Every boarding point is created and verified by a person who physically visits the location. The mandatory record is a name and a position; photographs, a night-safety flag and an accessibility flag are optional and encouraged.

This matters for a specific reason. A 2017 study in *PLOS ONE* measuring OpenStreetMap completeness worldwide identified Egypt among the least complete, estimating that fewer than one third of streets were mapped. The same study found that dense cities and main roads are the best-mapped parts of any country. Because boarding points are surveyed by hand and routes follow main corridors, the design depends very little on the weakest map data. Verifying this for the launch corridors is an engineering task listed in Chapter 7.

4 Revenue and cost mechanics

NO PROJECTIONS

This chapter describes how money is intended to move. It contains no forecast of volume, revenue or profit. The service has never operated, so any such figure would be invention.

4.1 Where revenue comes from

Commission	A percentage of each fare. The default model. Rate is a configuration value set per city, route or driver segment.
Subscriptions	Weekly or monthly commuter passes with a guaranteed seat. Predictable revenue and the strongest retention mechanism.
Upgraded boarding	A higher-priced ticket for collection from the passenger's own street rather than a stop.
Alternative models	Driver subscription, or a hybrid, are supported as configuration and can be enabled without engineering work.

4.2 How a fare is set

The stop-boarding fare is one flat figure per route. The street-collection fare is calculated in three layers: a fixed uplift, a charge banded by the size of the detour, and a cost-recovery component applied only if the first two do not cover the measured cost. The result is clamped between a floor and a ceiling, and it is shown to the passenger before they commit.

Beyond a defined detour threshold, street collection is not expensive — it is unavailable. A deviation that would delay the passengers already aboard is refused rather than priced.

4.3 Payment

Wallet balance is the primary method. Card, mobile wallet, kiosk cash-in, bank transfer and Apple Pay are supported for topping up or for direct payment, through a single payment provider integration.

Cash is accepted, with one rule: it is always recorded against the scanned booking. The driver marks the fare as collected, the amount becomes a liability owed by that driver to the platform, and it is deducted from their next payout. There is no untracked cash sale. A driver whose outstanding cash liability exceeds a configured limit is automatically prevented from accepting further cash bookings, having been warned in advance.

4.4 Cost structure

Routing and mapping	Self-hosted routing on open map data. Published comparisons put a self-hosted engine at roughly \$20 per month against roughly \$510 per day for equivalent commercial matrix queries at scale. One team reported moving from about \$8,000 per month to about \$520.
Infrastructure	Self-managed servers rather than managed cloud services, with a database replica and off-site backups.
Payment processing	Per-transaction and per-payout fees.
Messaging	Verification codes by SMS. All other messages are in-application, which avoids per-message cost.
Field operations	Surveying and verifying boarding points, and reviewing driver documents. Labour, not software.

NOT YET QUOTED

Infrastructure figures above are drawn from published third-party comparisons, not from quotations obtained for this project. Actual costs have not been established.

5 Technology

5.1 Shape of the system

One backend application, internally divided into sixteen modules with enforced boundaries, so that any module can later be separated into its own service without rewriting the parts that call it. A mobile application for passengers and drivers, and a web application for the public site and the operating dashboards.

Everything is written in one language across the applications and the server, with a single shared definition of each business object. A single Python service handles overnight optimisation.

5.2 Where decisions are made

GOVERNING RULE

If it affects money, safety, or who travels with whom, the server decides. If it affects only what one person sees on their own screen, the device may decide.

Client software can be modified and location data can be falsified, so every allocation and every charge is computed centrally. The device handles map display, location smoothing, local alarms and an offline queue.

5.3 Working without a connection

The driver application treats its own local storage as authoritative. A driver can complete an entire journey with no signal: boarding scans, arrival marks and cash records are stored locally and transmitted when connectivity returns, each carrying a key that prevents duplication.

Booking and payment remain online-only, because a seat cannot honestly be promised without checking that it is free.

5.4 Matching

Assignment runs as a six-stage pipeline. A geographic filter narrows candidates, a feasibility gate removes anything impossible or harmful, an insertion heuristic always produces a valid answer within milliseconds, and an optimiser improves on that answer within a fixed deadline. If the optimiser is interrupted at any moment, the earlier answer remains valid. No passenger ever waits on a solver.

Two findings shaped this. First, published measurements show that collecting requests for a short window before matching them together substantially outperforms matching each request instantly: reported average total waiting fell from roughly 609 seconds to roughly 427 seconds, and detour delay from roughly 190 to roughly 75 seconds. Second, a study using Munich trip data found an advanced matcher served up to 8 % more requests and saved 10 % more distance than a simple heuristic, but exceeded real-time limits as the problem grew — which is why the fast answer is always kept.

5.5 The fairness rule

Because passengers do not declare a destination, the system cannot promise an individual arrival time. Instead it protects the published timetable: no collection, deviation or wait may put the vehicle more than a defined margin behind schedule. Every passenger on a route shares one schedule, so protecting the schedule protects all of them.

5.6 Configuration rather than code

Fares, detour limits, waiting times, reward values, batch windows and service areas are stored as data and changed from the dashboard. Sixty-five parameters are catalogued with type, permitted range, default, who may change them, and — importantly — what the system does if the value is missing or unreadable. Published guidance on runtime configuration notes that many incidents occur during fallback rather than during the change itself, so each parameter declares its own safe behaviour: cash defaults to disabled, refund limits default to zero, and safety categories default to treating everything as severe.

Opening a new route, changing a price or launching a promotion requires no software release. The same mechanism is what allows a new city to be added as configuration.

5.7 Verification

A single command runs type checking, linting, module-boundary checks, unit tests, integration tests, contract tests, a build of every application, a secret scan and database migration checks. Nothing is merged unless it passes. Money handling carries the strictest testing requirement, and monetary values are stored as integers, never as floating-point numbers.

6 Trust, safety and governance

6.1 Verification of drivers

Every driver is reviewed by a person before carrying anyone: national identity document, driving licence, vehicle licence, a photograph checked against the identity document, and vehicle photographs. Automation may pre-check image quality and expiry dates, but never grants approval. Document expiry is monitored, with warnings at thirty, fourteen, seven and one day, and automatic suspension on expiry.

6.2 In-journey safety

An emergency control is reachable in one action from any journey screen and sends identity, live position, vehicle and driver details to an on-call responder. It can be triggered silently, in which case the system deliberately makes no sound — an audible alert could place the passenger in greater danger. A journey can be shared as a live link with a family member, and passengers and drivers can contact each other without either seeing the other’s telephone number.

6.3 How a driver is removed

Ratings are recorded but carry no automatic consequence. Removal happens through the complaint process, which is therefore built to carry that weight:

Serious categories	Immediate precautionary suspension pending investigation. The person stops driving first and is investigated second.
Complaint rate	A review is triggered at a complaint rate of 10 % of completed journeys, subject to a minimum of twenty journeys, measured over the last hundred.
Repeated pattern	Three complaints in the same category within thirty days trigger review even below that rate.
Human decision	No automatic removal in any circumstance. A person decides, records a reason, and the driver may appeal.

The 10 % figure was checked against published practice. Major platforms deactivate on a rating average of roughly 4.6 out of 5, which corresponds to approximately 8–12 % of passengers rating poorly. The minimum-volume and rolling-window conditions exist because a percentage alone punishes new drivers for a single complaint.

6.4 Money controls

Every movement of value is a double-entry record in one append-only ledger. Entries are never edited; corrections are new compensating entries. Support staff can cancel a journey and issue a refund up to one fare, three times a day, and cannot see identity documents or alter balances. Managers set prices but cannot move money. Two-factor authentication is mandatory for all staff. Promotional budgets are enforced in the ledger, so a campaign cannot overspend its allocation.

6.5 Data

All operational data is retained, with older records moved to lower-cost storage rather than deleted. Movement history is held separately with its own encryption, analytical access is pseudonymised by default, and re-identification is a separate action restricted to the highest administrative role and recorded. Bulk export requires approval and is logged.

REFERRED TO COUNSEL

Indefinite retention of personal movement data, and the regulatory position of the service under Egyptian transport law, are matters for legal counsel. They are recorded in the specification as open items and are not assessed here.

7 What is not known

Presenting a design without stating its uncertainties would misrepresent it. This chapter lists what has not been established.

7.1 Not yet measured

Corridor demand	No measurement of how many people travel which corridors at which hours. This determines whether the service works at all.
Map data quality	Egypt ranks among the least complete in open map coverage. The launch corridors must be surveyed before committing to self-hosted routing.
Operating parameters	Waiting times, detour limits, batch windows and walking distances have starting values only. Simulation against synthetic demand is planned before launch.
Infrastructure cost	No quotations obtained.
Willingness to pay	No pricing research with Alexandrian commuters has been conducted.

7.2 Principal risks

Risk	Description	Position
Regulatory	An open public transport service falls within Egypt's ride-hailing regulation, which carries substantial licensing obligations.	Referred to legal counsel.
Demand density	Shared transport fails below a demand threshold. Launching across a wide area with thin demand is the recognised failure pattern.	Mitigated by launching on a single corridor.
Driver supply	Matching optimises purely for efficiency, with no weighting toward drivers earning less. Industry practice usually includes such weighting because driver retention is the binding constraint.	A fairness weight exists, set to zero, and can be enabled without engineering work.
Coverage	If drivers do not claim departures, the timetable becomes unreliable and subscriptions cannot be honoured.	Subscriptions may only be sold against departures with committed coverage.
Cancellation policy	The intended penalty is stricter than the local incumbent, which offers a full refund up to three hours before travel.	Accepted, with mandatory disclosure before purchase and monitoring after launch.

Risk	Description	Position
Account sharing	Driver identity is verified once, at registration, with no periodic re-check. An unverified person driving on an approved account is the most serious failure the service could have.	Accepted. Mitigated by strict registration, the complaint process and anomaly detection. Recorded for monitoring.
Message delivery	Notifications are sent in-application only, with no SMS fallback. On a poor connection a passenger may not receive an arrival alert.	Mitigated: the arrival alarm fires from cached schedule data without requiring a live message. Delivery failure will be measured.
Team capacity	The specification is complete; construction has not begun. Delivery depends on resourcing that is outside this document.	Owned by the founder.

7.3 Deliberate omissions

No revenue, user or growth figures appear in this document because the service has not operated. No market size estimate is given because the underlying measurement has not been done. No timeline or hiring plan is included, as those decisions sit with the founder. No valuation or funding terms are proposed here.

8 Route to launch

The plan is sequential: one corridor working completely before widening. Each phase ends at a demonstrable condition rather than a date.

	Phase	Ends when
0	Foundations	The repository, automated checks, database, configuration system and audit trail exist and are deployable.
1	Identity and vehicles	A real driver can be verified and approved by a real administrator.
2	Geography	The launch corridor is fully surveyed and its boarding points verified.
3	The core journey	A passenger books, walks to a stop, boards by code and pays — on one corridor.
4	Safety and support	A passenger in difficulty reaches a person, and an unsafe driver can be removed.
5	Commercial control	A manager can change a price and run a promotion without engineering involvement.
6	Recurring travel	A commuter buys a monthly subscription and travels without further action.
7	Validation and launch	Simulation sets the operating parameters; a closed trial on one corridor meets defined criteria; public launch follows.

Widening beyond the first corridor is a dashboard action, not an engineering project. That is a direct consequence of holding all commercial values in configuration.

8.1 Before construction begins

1. Survey the launch corridor's map data and confirm the routing engine performs adequately on it.
2. Obtain infrastructure quotations.
3. Obtain a regulatory opinion from Egyptian counsel.
4. Commission the visual design from the existing design system and screen definitions.
5. Write the passenger-facing text in Arabic and English.

A Design decisions of record

The specification records 167 decisions, each with its reasoning and status. The following are those with the greatest bearing on how the service behaves commercially and operationally.

Ref	Decision	Effect
D-114	Passengers buy a route, not a destination	Board at a stop, alight anywhere along the route.
D-115	One flat fare per route	The same price for two stops or twelve.
D-113	No demand-based pricing	The price never rises with demand. A manager may set a temporary, attributable change.
D-132	Operator defines routes; drivers claim slots	Drivers carry no planning or pricing burden.
D-038	Boarding points created and verified by people	Every stop is physically visited before use.
D-140	Boarding fixed, alighting free	Collection must be predictable; setting down need not be.
D-056	Price fixed at purchase	The quoted fare never changes afterwards.
D-049	Boarding confirmed by code	Works without a network connection.
D-078	Cash always recorded against a booking	No untracked cash sale can exist.
D-070	Commercial values held as configuration	Prices, routes and promotions change without a software release.
D-031	No organisational accounts	Universities are a sales channel; everyone uses one open network.
D-051	Subscriptions with a guaranteed seat	Recurring revenue and the principal retention mechanism.
D-096	Ratings carry no automatic consequence	Removal happens through the complaint process, under human decision.
D-138	Complaint-rate review threshold	10 % over a hundred journeys, minimum twenty, triggering human review.
D-110	One corridor complete, then widen	Concentrates demand where shared transport requires density.

B Sources

Every external figure in this document is listed here. Nothing quantitative is presented without a source, and no figure originates from this project's own operation, because it has none.

Fixed-route versus demand-responsive transit. *Fixed-route or demand-responsive transit? An evaluation framework for transit service structures using dual-perspective indicators*, ScienceDirect, 2026. Includes a comparative table of fourteen studies published 2007–2025. Source of the finding that fixed-route services perform better on cost, schedule adherence and ridership under dense, regular demand.

Cost per passenger comparison. CivicWell, reporting AC Transit's flexible-service pilot: approximately \$72 per passenger against approximately \$25 on the fixed route it replaced. A cited review of rural England reports £1.35–1.62 for fixed route against £9–19.96 for demand-responsive.

When not to use demand-responsive transit. Movmi, *What Is DRT? Demand-Responsive Transit Explained*. Lists dense urban routes and peak-hour commuter corridors among the conditions where demand-responsive services underperform.

Discount sustainable by walk-to-a-point services. Getridewise, 2026 comparison of shared-ride products: door-to-door shared services measured at 15–22 % below a private ride; walk-to-corner services at 25–40 %. Contemporary reporting of Uber Express Pool and Lyft Shared Saver corroborates the pattern.

Effect of batching ride requests. *Timing the Match: A Deep Reinforcement Learning Approach for Ride-Hailing and Ride-Pooling Services*, arXiv 2503.13200. Reported average total waiting of 608.8 seconds under immediate dispatch against 427.0 seconds with a twenty-second batching window; detour delay 190.5 against 113.1 seconds.

Heuristic versus optimiser in ride-pooling. Engelhardt et al., *Speed-up Heuristic for an On-Demand Ride-Pooling Algorithm*, arXiv 2007.14877. Using Munich data: an advanced matcher served up to 8 % more requests and saved 10 % more distance, but exceeded real-time computation limits as problem size grew.

High-capacity ride-sharing algorithm. Alonso-Mora et al., *On-demand high-capacity ride-sharing via dynamic trip-vehicle assignment*, PNAS, 2017. Source of the anytime-optimal approach adopted in the matching design.

Open map data completeness. Barrington-Leigh and Millard-Ball, *The world's user-generated road map is more than 80% complete*, PLOS ONE, 2017. Estimates fewer than one third of streets mapped in Egypt, and finds dense cities and main roads best mapped.

Self-hosted routing cost. Published benchmark comparing self-hosted routing engines against commercial matrix services: approximately 21 milliseconds per 100×100 matrix self-hosted, against 2,500 milliseconds and approximately \$510 per day commercially. A practitioner report describes reducing mapping spend from approximately \$8,000 to approximately \$520 per month.

Comparable operator. Swvl Holdings Corp. Founded Cairo 2017; listed 2022; moved away from consumer mass-market operation; reported profitability in 2025 on business and government contracts with a high proportion of recurring revenue.

Ride-hailing regulation in Egypt. Egyptian ride-hailing law passed 2018 with implementing regulations in 2019 (Law 2180/2019). Reported obligations include a five-year operating licence,

individual driver permits, vehicle standards and trip-data retention. Referred to legal counsel; not assessed in this document.

Driver deactivation thresholds. Published summaries of major platform practice indicate deactivation at a rating average of approximately 4.6 out of 5, and that repeated complaints of the same type are acted on independently of the average.

Usability of existing applications. Hsu and Chen, *Usability Study on the User Interface Design of Ride-hailing Applications*, Springer, 2023. System Usability Scale scores across thirty participants: Uber 66.75, Lyft 60.25, Gojek 62.75, against a conventional average of 68.

Notification practice. Published 2026 guidance on notification tiers and frequency limits, and Nielsen Norman Group on transactional notifications, including the finding that push delivery is connectivity-dependent and may fail in poor-signal environments.

Runtime configuration practice. Published guidance on runtime configuration change management, including the observation that many incidents occur during fallback behaviour rather than during the change itself.

Payment infrastructure in Egypt. Central Bank of Egypt materials on the Instant Payment Network, launched March 2022 and operated by the Egyptian Banks Company across thirty-eight member banks; Paymob developer documentation for collection and disbursement.

Where a source is described as reported or published, it has not been independently verified by the author. Items referred to legal counsel are outside the scope of this document.

C Supporting documents

This document summarises a larger body of work, available on request.

Document	Contents
Master Specification v3.0	The complete build document. Twenty-two chapters covering the domain model, roles, journey lifecycle, geography, supply model, matching algorithm, money and ledger, pricing, growth mechanics, dashboards, system and code architecture, data model and interface contract, design system, seventy-five screen definitions, safety, privacy, quality, infrastructure, operations runbook, configuration catalogue and message catalogue. Approximately 60,000 words.
Questions and Answers	All 167 decisions with the question asked, the answer given, the reasoning and the status, including an appendix of superseded decisions.
Research findings	Eighteen research passes with full citations, covering regulation, comparable operators, matching algorithms, pricing evidence, mapping infrastructure, payment rails, usability and configuration practice.
Item tracker	Fifty-six tracked items with their disposition: closed, specified, monitored, delegated or outstanding.

End of document